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ProjExD_Group14 / koukaton_up.py 

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 c0a25046bf コード最終版

7f7f5a0 · 1 minute ago 

563 lines (458 loc) · 21.9 KB

Code Blame  Raw 

```
1  import pygame
2  import sys
3  import os
4  import random
5
6  # カレントディレクトリの固定
7  os.chdir(os.path.dirname(os.path.abspath(__file__)))
8
9  # =====
10 # 設計方針・共通ルール
11 # 1. クラス内の変数は、すべて「get_変数名」というメソッドを介してアクセスする
12 # 2. クラスに関係するすべての操作関数は、クラスの外部で定義する
13 # =====
14
15 # 画面・物理設定
16 WIDTH, HEIGHT = 600, 800
17 FPS = 60
18 GRAVITY = 0.6
19
20 # カラー定義
21 BLACK = (20, 20, 25)
22 BLUE = (100, 150, 255)
23 WHITE = (255, 255, 255)
24 GOLD = (255, 215, 0)
25
26 DARK_PANEL = (35, 35, 45)
27 PANEL_BORDER = (90, 90, 110)
28 LIGHT_BLUE = (120, 200, 255) # ステータスUI用 (青)
29 ORANGE = (255, 170, 80) # ベスト記録UI用 (オレンジ)
30
31 # ギミック床の色定義
32 COLOR_NORMAL = (120, 120, 120) # 通常の床 (灰色)
33 COLOR_FAKE = (180, 0, 180) # すり抜ける床 (紫)
34 COLOR_ICE = (100, 200, 255) # 滑る床 (水色)
35 COLOR_TRAMPOLINE = (255, 140, 0) # 大ジャンプする床 (オレンジ)
36 COLOR_TRAP = (255, 50, 50) # 最初に戻る床 (赤)
37 COLOR_RED_UI = (255, 100, 100) # チャージゲージ用 (赤)
38
39 # 種類から色を取得するマッピング辞書
40 COLOR_MAP = {
41     "normal": COLOR_NORMAL,
42     "fake": COLOR_FAKE,
```

```
43     "ice": COLOR_ICE,
44     "trampoline": COLOR_TRAMPOLINE,
45     "trap": COLOR_TRAP
46 }
47
48 # =====
49 # 1. データ構造（クラス）の定義
50 # =====
51
52 class Player:
53     def __init__(self):
54         # 画像・位置情報
55         self.image_original = pygame.image.load("9.png")
56         self.image_original = pygame.transform.scale(self.image_original, (30, 40))
57         self._image = self.image_original
58         self._rect = self._image.get_rect(center=(WIDTH // 2, HEIGHT - 100))
59
60         # 移動・物理状態
61         self._vel_x = 0
62         self._vel_y = 0
63         self._on_ground = False
64
65         # ジャンプチャージ設定
66         self._is_charging = False
67         self._charge_power = 0
68         self._max_charge = 20.0
69         self._direction = 1          # 1: 右, -1: 左
70
71         # ゲーム状態
72         self._is_clear = False
73         self._on_ice = False
74         self._is_reset = False
75
76         # クリエイティブモード設定
77         self._is_creative = False
78         self._has_cheated = False   # クリエイティブ使用履歴（記録更新不可フラグ）
79
80     # --- ゲッターメソッド ---
81     def get_image(self): return self._image
82     def get_rect(self): return self._rect
83     def get_vel_x(self): return self._vel_x
84     def get_vel_y(self): return self._vel_y
85     def get_on_ground(self): return self._on_ground
86     def get_is_charging(self): return self._is_charging
87     def get_charge_power(self): return self._charge_power
88     def get_max_charge(self): return self._max_charge
89     def get_direction(self): return self._direction
90     def get_is_clear(self): return self._is_clear
91     def get_on_ice(self): return self._on_ice
92     def get_is_reset(self): return self._is_reset
93     def get_is_creative(self): return self._is_creative
94     def get_has_cheated(self): return self._has_cheated
95
96     # --- セッターメソッド ---
97     def set_vel_x(self, value): self._vel_x = value
98     def set_vel_y(self, value): self._vel_y = value
99     def set_on_ground(self, value): self._on_ground = value
100    def set_is_charging(self, value): self._is_charging = value
```

```
101     def set_charge_power(self, value): self._charge_power = value
102     def set_direction(self, value): self._direction = value
103     def set_is_clear(self, value): self._is_clear = value
104     def set_on_ice(self, value): self._on_ice = value
105     def set_is_reset(self, value): self._is_reset = value
106     def set_is_creative(self, value): self._is_creative = value
107     def set_has_cheated(self, value): self._has_cheated = value
108
109
110     class Platform:
111     def __init__(self, x, y, w, h, color, plat_type="normal"):
112         self._image = pygame.Surface((w, h))
113         self._image.fill(color)
114         self._rect = self._image.get_rect(topleft=(x, y))
115         self._type = plat_type
116
117         # --- ゲッターメソッド ---
118         def get_image(self): return self._image
119         def get_rect(self): return self._rect
120         def get_type(self): return self._type
121
122
123     # =====
124     # 2. クラスを操作する関数（クラス外部定義）
125     # =====
126
127     def update_player(player, keys, platforms, goal_block, jump_sound):
128         """プレイヤーの移動、入力、衝突判定などの状態を更新する関数"""
129         if not player.get_is_clear():
130
131             # --- クリエイティブモードの切り替え制御 ---
132             if keys[pygame.K_LSHIFT] and keys[pygame.K_c]:
133                 player.set_is_creative(True)
134                 player.set_has_cheated(True) # チート履歴をON（トラップを踏むまで記録更新をストップ）
135                 player.set_vel_x(0)
136                 player.set_vel_y(0)
137                 player.set_is_charging(False)
138                 player.set_charge_power(0)
139             elif keys[pygame.K_LSHIFT] and keys[pygame.K_a]:
140                 player.set_is_creative(False)
141
142             # --- クリエイティブモード中の自由移動処理 ---
143             if player.get_is_creative():
144                 speed = 10
145                 rect = player.get_rect()
146
147                 if keys[pygame.K_LEFT] or keys[pygame.K_a]:
148                     rect.x -= speed
149                     player.set_direction(-1)
150                     player._image = pygame.transform.flip(player.image_original, True, False)
151                 if keys[pygame.K_RIGHT] or keys[pygame.K_d]:
152                     rect.x += speed
153                     player.set_direction(1)
154                     player._image = player.image_original
155                 if keys[pygame.K_UP] or keys[pygame.K_w]:
156                     rect.y -= speed
157                 if keys[pygame.K_DOWN] or keys[pygame.K_s]:
158                     rect.y += speed
```

```
159
160     # 画面左右端の移動制限
161     if rect.left < 0: rect.left = 0
162     if rect.right > WIDTH: rect.right = WIDTH
163
164     return # クリエイティブ中は物理演算や通常の当たり判定をスキップ
165
166     # --- 通常モード：地上での移動とチャージ処理 ---
167     if player.get_on_ground():
168         if player.get_on_ice():
169             player.set_vel_x(player.get_vel_x() * 0.96) # 氷の上は徐々に減速（滑る）
170         else:
171             player.set_vel_x(0)
172
173     # 向きの変更
174     if keys[pygame.K_LEFT] or keys[pygame.K_a]:
175         player.set_direction(-1)
176         player._image = pygame.transform.flip(player.image_original, True, False)
177     if keys[pygame.K_RIGHT] or keys[pygame.K_d]:
178         player.set_direction(1)
179         player._image = player.image_original
180
181     # ジャンプチャージ
182     if keys[pygame.K_SPACE]:
183         player.set_is_charging(True)
184         current_power = player.get_charge_power()
185         if current_power < player.get_max_charge():
186             player.set_charge_power(current_power + 0.4)
187     else:
188         # スペースキーが離されたらジャンプを実行
189         if player.get_is_charging():
190             power = player.get_charge_power()
191             direction = player.get_direction()
192
193             player.set_vel_y(-power * 0.9 - 5)
194             player.set_vel_x(direction * (power * 0.4 + 2))
195
196             if jump_sound:
197                 jump_sound.play()
198
199             player.set_is_charging(False)
200             player.set_charge_power(0)
201             player.set_on_ground(False)
202             player.set_on_ice(False)
203
204     # 空中にいる場合は重力を適用
205     if not player.get_on_ground():
206         player.set_vel_y(player.get_vel_y() + GRAVITY)
207
208     rect = player.get_rect()
209
210     # --- x軸方向の移動と壁・足場の側面衝突判定 ---
211     rect.x += player.get_vel_x()
212     if rect.left < 0:
213         rect.left = 0
214         player.set_vel_x(player.get_vel_x() * -0.5) # 跳ね返り
215         player.set_direction(1)
216     if rect.right > WIDTH:
```

```
117         rect.right = WIDTH
118         player.set_vel_x(player.get_vel_x() * -0.5) # 跳ね返り
119         player.set_direction(-1)
120
121     for plat in platforms:
122         if plat.get_type() == "fake":
123             continue # 紫の床 (fake) は側面判定もスルー
124
125         plat_rect = plat.get_rect()
126         if rect.colliderect(plat_rect):
127             if player.get_vel_x() > 0: # 右移動中に衝突
128                 rect.right = plat_rect.left
129                 player.set_vel_x(player.get_vel_x() * -0.5)
130                 player.set_direction(-1)
131             elif player.get_vel_x() < 0: # 左移動中に衝突
132                 rect.left = plat_rect.right
133                 player.set_vel_x(player.get_vel_x() * -0.5)
134                 player.set_direction(1)
135
136     # --- Y軸方向の移動と天井・着地判定 ---
137     rect.y += int(player.get_vel_y())
138
139     # ゴールブロックとの衝突判定 (通常モード時のみクリア可能)
140     goal_rect = goal_block.get_rect()
141     if rect.colliderect(goal_rect):
142         if player.get_vel_y() < 0:
143             rect.top = goal_rect.bottom
144             player.set_vel_y(0)
145             player.set_is_clear(True)
146             pygame.mixer.music.fadeout(2000) # BGMをフェードアウト
147
148     player.set_on_ground(False)
149     player.set_on_ice(False)
150
151     # 通常の足場に対する上下の衝突判定ループ
152     for plat in platforms:
153         if plat.get_type() == "fake":
154             continue # 紫の床 (fake) は上下判定も完全にスルー
155
156         plat_rect = plat.get_rect()
157         if rect.colliderect(plat_rect):
158             # 落下中の着地判定
159             if player.get_vel_y() > 0:
160                 if rect.bottom <= plat_rect.top + player.get_vel_y() + 1:
161
162                     # トランポリン床 (オレンジ)
163                     if plat.get_type() == "trampoline":
164                         rect.bottom = plat_rect.top
165                         player.set_vel_y(-25)
166                         player.set_on_ground(False)
167                         break
168
169                     # トラップ床 (赤)
170                     if plat.get_type() == "trap":
171                         rect.center = (WIDTH // 2, HEIGHT - 100)
172                         player.set_vel_y(0)
173                         player.set_vel_x(0)
174                         player.set_is_reset(True)
```

```
275         player.set_has_cheated(False) # スタートに戻ったらペナルティを解除
276         break
277
278     # 通常の着地処理
279     rect.bottom = plat_rect.top
280     player.set_vel_y(0)
281
282     if plat.get_type() == "ice":
283         player.set_on_ice(True)
284     else:
285         player.set_on_ice(False)
286
287     player.set_on_ground(True)
288     break
289
290     # 上昇中に天井（足場の裏側）に頭をぶつけた判定
291     elif player.get_vel_y() < 0:
292         rect.top = plat_rect.bottom
293         player.set_vel_y(0)
294         break
295
296
297 ✓ def draw_ui(screen, font, player, current_floor, total_floors, current_height, max_height):
298     """画面左上のステータスUIを描画する関数"""
299     panel_x = 10
300     panel_y = 10
301     panel_w = 300
302     panel_h = 120
303
304     # UIの枠組み
305     panel_rect = pygame.Rect(panel_x, panel_y, panel_w, panel_h)
306     pygame.draw.rect(screen, DARK_PANEL, panel_rect, border_radius=12)
307     pygame.draw.rect(screen, PANEL_BORDER, panel_rect, 2, border_radius=12)
308
309     # テキストレンダリングと配置
310     title_text = font.render("STATUS", True, GOLD)
311     screen.blit(title_text, (panel_x + 15, panel_y + 10))
312
313     floor_text = font.render(f"Floor {current_floor} / {total_floors}", True, WHITE)
314     screen.blit(floor_text, (panel_x + 15, panel_y + 42))
315
316     height_text = font.render(f"Height: {current_height // 10} m", True, LIGHT_BLUE)
317     screen.blit(height_text, (panel_x + 15, panel_y + 70))
318
319     # クリエイティブ使用中はベスト記録の色を変更（更新停止中を表現）
320     max_color = ORANGE if not player.get_has_cheated() else (150, 150, 150)
321     max_text = font.render(f"Best: {max_height // 10} m", True, max_color)
322     screen.blit(max_text, (panel_x + 180, panel_y + 70))
323
324     # クリエイティブモード稼働中のテキスト表示
325     if player.get_is_creative():
326         creative_text = font.render("[ CREATIVE MODE ]", True, (255, 100, 255))
327         screen.blit(creative_text, (panel_x + 5, panel_y + panel_h + 5))
328
329
330     # =====
331     # 3. メインループ
332     # =====
```

```
333
334 ▼ def main():
335     pygame.init()
336     pygame.mixer.init()
337
338     screen = pygame.display.set_mode((WIDTH, HEIGHT))
339     pygame.display.set_caption("koukaton UP!!")
340     clock = pygame.time.Clock()
341     font = pygame.font.SysFont(None, 36)
342     large_font = pygame.font.SysFont(None, 72)
343
344     # サウンドの読み込み
345     jump_sound = None
346     if os.path.exists("jump.wav"):
347         jump_sound = pygame.mixer.Sound("jump.wav")
348         jump_sound.set_volume(0.4)
349
350     if os.path.exists("BGM.mp3"):
351         pygame.mixer.music.load("BGM.mp3")
352         pygame.mixer.music.set_volume(0.25)
353         pygame.mixer.music.play(-1)
354
355     player = Player()
356
357     # 背景画像の読み込みとサイズ調整
358     bg_large = pygame.image.load("background_all.png")
359     bg_large = pygame.transform.scale(bg_large, (600, 2400))
360
361     TOTAL_FLOORS = 10
362     platforms = []
363
364     # 初期位置の床 (一番下)
365     platforms.append(Platform(0, HEIGHT - 40, WIDTH, 40, COLOR_NORMAL, "normal"))
366
367     # 固定ステージデータ (下から順にすべての床の番号をコメントアウトで記載)
368     # 形式: (x座標, y座標, 幅, ギミック種類)
369     fixed_level_data = [
370         # Floor 1 (Easy - 広くて近い)
371         (100, 600, 150, "normal"), # 床1
372         (400, 450, 150, "normal"), # 床2
373         (150, 300, 150, "normal"), # 床3
374         (350, 150, 150, "normal"), # 床4
375         (250, 0, 150, "normal"), # 床5
376
377         # Floor 2 (Easy - 少し狭くなる)
378         (50, -150, 120, "normal"), # 床6
379         (400, -300, 120, "normal"), # 床7
380         (100, -450, 120, "normal"), # 床8
381         (350, -600, 120, "normal"), # 床9
382         (200, -750, 120, "normal"), # 床10
383
384         # Floor 3 (氷の床 初登場)
385         (450, -900, 100, "normal"), # 床11
386         (150, -1050, 100, "ice"), # 床12
387         (400, -1200, 100, "normal"), # 床13
388         (200, -1350, 100, "ice"), # 床14
389         (100, -1500, 100, "normal"), # 床15
390
```

```
391         # Floor 4 (トランポリン 初登場)
392         (350, -1650, 100, "trampoline"), # 床16
393         (100, -2100, 120, "normal"), # 床17
394         (400, -2250, 100, "normal"), # 床18
395         (200, -2400, 100, "normal"), # 床19
396
397         # Floor 5 (すり抜ける罠 初登場)
398         (50, -2550, 100, "normal"), # 床20
399         (300, -2600, 100, "fake"), # 床21
400         (400, -2700, 100, "normal"), # 床22
401         (150, -2850, 100, "ice"), # 床23
402         (50, -3000, 100, "fake"), # 床24
403         (350, -3050, 100, "normal"), # 床25
404         (100, -3200, 100, "normal"), # 床26
405
406         # Floor 6 (幅が狭くなる)
407         (450, -3350, 80, "normal"), # 床27
408         (200, -3500, 80, "ice"), # 床28
409         (50, -3650, 80, "trampoline"), # 床29
410         (350, -4100, 80, "normal"), # 床30
411
412         # Floor 7 (トラップ床 初登場)
413         (150, -4250, 80, "normal"), # 床31
414         (400, -4400, 80, "trap"), # 床32
415         (450, -4500, 80, "normal"), # 床33
416         (100, -4650, 80, "normal"), # 床34
417         (250, -4800, 80, "ice"), # 床35
418
419         # Floor 8 (複合ギミック)
420         (450, -4950, 70, "normal"), # 床36
421         (50, -5100, 70, "fake"), # 床37
422         (150, -5200, 70, "normal"), # 床38
423         (300, -5350, 70, "trap"), # 床39
424         (50, -5450, 70, "normal"), # 床40
425         (400, -5600, 70, "trampoline"), # 床41
426
427         # 41番目と42番目の間の追加床 (左寄り)
428         (80, -5850, 70, "normal"), # 床42
429
430         # Floor 9 (終盤・かなり狭い)
431         (75, -6050, 70, "normal"), # 床43
432         (400, -6200, 70, "ice"), # 床44
433         (150, -6350, 70, "trap"), # 床45
434         (200, -6450, 70, "normal"), # 床46
435
436         # Floor 10 (最後の関門)
437         (450, -6600, 60, "normal"), # 床47
438         (250, -6750, 60, "ice"), # 床48
439         (50, -6900, 60, "trampoline"), # 床49
440     ]
441
442     # 固定ステージデータを読み込んでリスト化
443     for data in fixed_level_data:
444         x, y, w, plat_type = data
445         platforms.append(Platform(x, y, w, 20, COLOR_MAP[plat_type], plat_type))
446
447     # ゴールブロックの設定
448     goal_y = (HEIGHT - 40) - (HEIGHT * TOTAL_FLOORS)
```



```
449     goal_block = Platform(0, goal_y, WIDTH, 40, GOLD, "goal")
450
451     camera_y = 0
452     max_height = 0
453     clear_time = None # クリアした瞬間のゲームティック記録用
454
455     running = True
456     while running:
457         for event in pygame.event.get():
458             if event.type == pygame.QUIT:
459                 running = False
460
461         keys = pygame.key.get_pressed()
462
463         # プレイヤー状態のアップデート
464         update_player(player, keys, platforms, goal_block, jump_sound)
465
466         p_rect = player.get_rect()
467
468         # 赤いトラップ床を踏んでリセットされた場合のカメラ同期
469         if player.get_is_reset():
470             camera_y = 0
471             player.set_is_reset(False)
472
473         # カメラの上方向スクロール処理
474         SCROLL_TOP_MARGIN = 200
475         if p_rect.y - camera_y < SCROLL_TOP_MARGIN:
476             camera_y = p_rect.y - SCROLL_TOP_MARGIN
477
478         # カメラの下方向滑らかな追従スクロール処理
479         SCROLL_BOTTOM_MARGIN = 150
480         camera_easing_down = 0.08
481         if p_rect.bottom - camera_y > HEIGHT - SCROLL_BOTTOM_MARGIN:
482             target_camera_y = p_rect.bottom - (HEIGHT - SCROLL_BOTTOM_MARGIN)
483             camera_y += (target_camera_y - camera_y) * camera_easing_down
484
485         if camera_y > 0:
486             camera_y = 0
487
488         # 高さ計算とベスト記録の保存（クリエイティブ使用中は記録更新をブロック）
489         current_height = (HEIGHT - 40) - p_rect.bottom
490         if current_height > max_height and not player.get_has_cheated():
491             max_height = current_height
492
493         current_floor = min((current_height // HEIGHT) + 1, TOTAL_FLOORS)
494         display_floor = max(1, current_floor)
495
496         # 背景描画（パララックス効果付き）
497         bg_y = 1600 + (camera_y * 0.2)
498         if bg_y < 0: bg_y = 0
499         src_rect = pygame.Rect(0, int(bg_y), 600, 800)
500         screen.blit(bg_large, (0, 0), src_rect)
501
502         # ゴールブロックの描画
503         goal_rect = goal_block.get_rect()
504         goal_draw_y = goal_rect.y - camera_y
505         if -100 < goal_draw_y < HEIGHT + 100:
506             screen.blit(goal_block.get_image(), (goal_rect.x, goal_draw_y))
```

```
507
508     # 足場の描画
509     for plat in platforms:
510         plat_rect = plat.get_rect()
511         draw_y = plat_rect.y - camera_y
512         if -50 < draw_y < HEIGHT + 50:
513             screen.blit(plat.get_image(), (plat_rect.x, draw_y))
514
515     # プレイヤーの描画
516     player_draw_y = p_rect.y - camera_y
517     screen.blit(player.get_image(), (p_rect.x, player_draw_y))
518
519     # 通常モードかつジャンプチャージ中のみゲージを描画
520     if player.get_is_charging() and not player.get_is_clear() and not player.get_is_creative():
521         gauge_width = (player.get_charge_power() / player.get_max_charge()) * 40
522         pygame.draw.rect(screen, COLOR_RED_UI, (p_rect.centerx - 20, player_draw_y - 15, gauge_width, 8
523
524     # プレイヤーの向きを示すガイドドットの描画
525     arrow_x = p_rect.right + 5 if player.get_direction() == 1 else p_rect.left - 10
526     pygame.draw.rect(screen, WHITE, (arrow_x, player_draw_y + 15, 5, 5))
527
528     # 情報表示 (UIの描画)
529     draw_ui(screen, font, player, current_floor, TOTAL_FLOORS, current_height, max_height)
530
531     # クリア時の演出および10秒後の自動シャットダウン処理
532     if player.get_is_clear():
533         if clear_time is None:
534             clear_time = pygame.time.get_ticks()
535
536     # クリアテキスト描画
537     clear_text = large_font.render("CLEAR!!", True, GOLD)
538     text_rect = clear_text.get_rect(center=(WIDTH // 2, HEIGHT // 2 - 50))
539     screen.blit(clear_text, text_rect)
540
541     sub_text = font.render("CONGRATULATIONS!", True, WHITE)
542     sub_rect = sub_text.get_rect(center=(WIDTH // 2, HEIGHT // 2 + 10))
543     screen.blit(sub_text, sub_rect)
544
545     # 自動終了までの秒数カウントダウン表示
546     elapsed = pygame.time.get_ticks() - clear_time
547     remaining = max(0, 10 - (elapsed // 1000))
548     close_text = font.render(f"Closing in {remaining} seconds...", True, LIGHT_BLUE)
549     close_rect = close_text.get_rect(center=(WIDTH // 2, HEIGHT // 2 + 60))
550     screen.blit(close_text, close_rect)
551
552     # 10秒 (10000ミリ秒) 経過でループを終了
553     if elapsed >= 10000:
554         running = False
555
556     pygame.display.flip()
557     clock.tick(FPS)
558
559     pygame.quit()
560     sys.exit()
561
562 if __name__ == "__main__":
563     main()
```

